Tunable SOA Wavelength Converter for Optical Packet Switching Router

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***Abstract*— We examine wavelength conversion for a packet switching router with in-band labels using efficient and cascadable all-optical wavelength conversion (AOWC) via a low-cost dual-pump SOA scheme. System performance is examined over a 35nm tuning range.**

I. Introduction and Router Architecture

The expanding scale of computer networks and the push to resource virtualization requires reliable inter-connection at lower operational cost. Fiber optic networks with wavelength routing can provide such efficient high-speed interconnection for data centers (DCs) and will be a key enabling technology to implement these virtual computer networks.

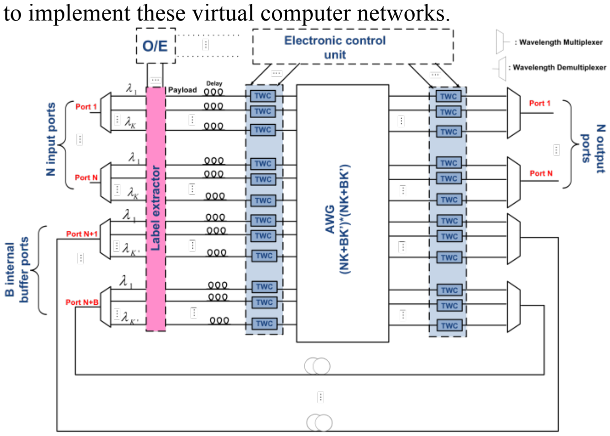

Fig. 1. Optical packet switching router architecture

Fig. 1 depicts a *(NK+BK')×(NK+BK')* switch architecture [1]: each of N input ports carries K wavelengths; each of B ports used for recirculation carries K' wavelengths that buffer contending packets in fiber delay-lines until the switch can forward them. The signals at each switch input port are demultiplexed into separate fibers. Each of these input channels is connected to an input port of the AWG via a tunable wavelength converter (TWC). Incoming signals contain routing information in an optical label; the label is extracted and used to determine the input and output TWC settings. We note that the input TWC requires a fixed wavelength input and variable wavelength output, whereas the output TWC requires a variable wavelength input and fixed wavelength output. A fixed wavelength laser can be amplified and shared by different TWCs, whereas one tunable wavelength laser should be dedicated to each TWC. TWC settings are determined by routing information carried by the label. Compared to the routing fabric proposed in [2], this router scheme offers more network management flexibility as any incoming wavelength at any port can be routed to a designated port with a desired wavelength. The main challenge in the router implementation is TWCs covering the C-band, offering low penalty, and being compatible with multiple modulation formats. Four wave mixing (FWM) in highly non-linear SOAs offers a practical AOWC solution for DC applications due to its polarization independency, compactness and low pump power requirement compared to other proposed technologies such as highly non-linear fiber (HNLF) and periodically poled $LiNbO_3$ (PPLN) [3].

II. Experimental Setup

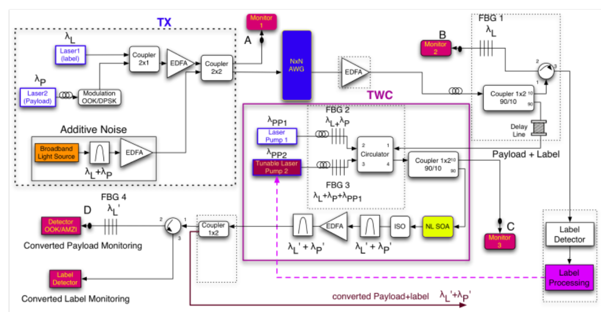

Fig. 2. Experimental SOA TWC using copolarized dump pumps

In Fig. 2 we show the dual-pump SOA TWC scheme. In the transmitter box, the label that contains routing information is encoded on wavelength λ_L separate but very close to the payload wavelength λ_P so that they can propagate through the same AWG port. This scheme does not require label rewriting and reinsertion as the label will always propagate with the payload in the same channel before it reaches its destination node, hence alleviates the router complexity and cost. A low noise EDFA is used to adjust the optical power before wavelength conversion. In the TWC box, Pump1 is set at a fixed wavelength λ_{PP1} close to λ_P and λ_L, while the wavelength λ_{PP2} of Pump2 is set close to the desired output wavelength according the information extracted from the label detection and processing box.

The payload and label are combined with Pump1 and Pump2 via a 4-port circulator and two fiber Bragg gratings (FBG) resulting in an incurred payload loss of about 2.2 dB, mostly due to the 3 times circulator port-to-port path loss of 0.7dB each. Pump2 λ_{PP2} can be tuned to any wavelength outside the band containing λ_L, λ_P, λ_{PP1}, (reflected by FBG3). All four wavelengths are then combined at output port 4. For AOWC we use a CIP NL-OEC SOA. The payload and the in-band label that propagate through the same AWG port are converted together to λ'_P and λ'_L via a dual-pump FWM process and then filtered. At the output of the SOA, we introduce a two-stage filters and an EDFA to amplify the converted signals at wavelengths at λ'_P and λ'_L and to remove the Pump1, Pump2 and ASE noise. Points A, B, C, and D are used for monitoring purposes.

	λ (nm)	OOK (dBm)	DPSK (dBm)
λ_P	1555.0	1.0	-1.5
λ_L	1554.7	-3.0	-5.5
λ_{PP1}	1555.4	4.4	5.5
λ_{PP2}	1570.0	4.5	5.5
λ'_P	1569.6	-12.5	-15.4
λ'_L	1569.3	-17.1	-18.0

Table 1. Typical Payload and Pump power for 10Gb/s OOK & DPSK signal

III. Results and Discussion

We examine the experimental TWC performance for both on-off keying (OOK) and differential phase-shift keying (DPSK) modulation format at 10Gb/s. Typical signal and pump powers at the NL SOA input can be found in Table 1.

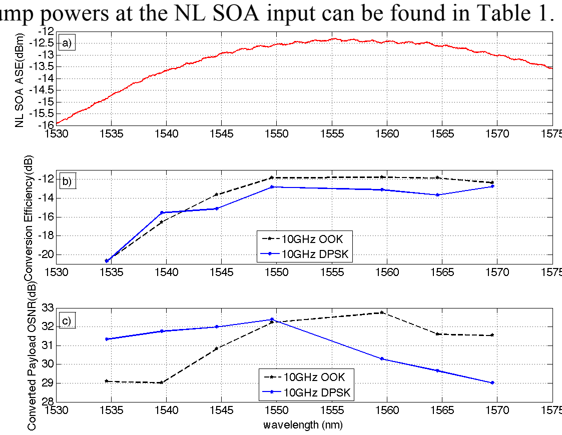

Fig. 3. TWC using Dual-pump FWM in SOA with input OSNR = 45dB

In Fig. 3b), we see that the conversion efficiency of 10 Gb/s OOK and NRZ-DPSK varied 7-8 dB across the SOA spectral bandwidth (displayed in Fig. 3a)), but the converted OSNR shown in Fig. 3c) experienced less variation (3-4 dB) as the ASE was suppressed by the strong pump power. We consider OSNR more critical than the payload conversion efficiency as poor efficiency can be compensated by amplification. Cascadability of the packet switching routers and buffers also require a low OSNR penalty for wavelength conversion.

Bit error rates (BER) were measured and are reported in Fig. 4. The payload contains 2^{31}-1 pseudo-random sequences at 1535-1570 nm with the same input powers. The converted payloads of both the OOK and DPSK signals were pre-amplified to assure 0 dBm at each wavelength; two 25 GHz pass-band filters were adjusted to maximize the converted payload output power. Error-free conversion can be achieved when the input OSNR is higher than 22 dB for OOK and 18 dB for DPSK. For OOK at BER = 10^{-9} we observe 1 dB OSNR penalty for $\lambda'_P = \lambda_P \pm 5$ nm, 3 dB ONSR penalty for $\lambda'_P = \lambda_P \pm 10$ nm and $\lambda'_P = \lambda_P \pm 15$ nm. For DPSK we used a single-ended detection scheme with delayed Mach-Zehnder interferometer, hence the original payload has BER performance very close to that of OOK. However from Fig. 4b) we observe less than 1 dB penalty at 10^{-9} over the entire C-band. Furthermore at +5 nm and +10 nm we observed improved BER performance.

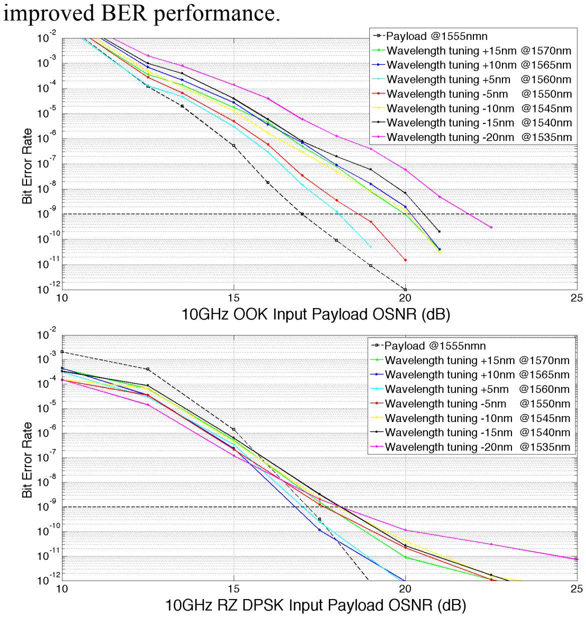

Fig. 4. a) 10Gb/s OOK, b) 10Gb/s DPSK BER after conversion

IV. Conclusions

We have demonstrated wavelength conversion supporting a versatile optical packet switching router design. The performance of 10 Gb/s OOK and DPSK was examined over the whole C-band with an in-band label. As a matter of fact, as the SOA operates in deep saturation region, its carrier density and gain can be modulated instantaneously by the input signal power and transferred to the converted signals, which explains why the constant envelop DPSK detection outperforms the OOK after conversion.

References

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